

AMRUTHA SHYAM

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EDUCATION

M.S. Computer Science, New York University, Tandon School of Engineering

Sep 2025 - May 2027

Relevant Coursework: Artificial Intelligence, ML, Responsible AI, Computer Vision, Information Visualization, Big Data

B.Tech Computer Science (AI & ML), Vellore Institute of Technology (VIT), Chennai

Sep 2021 - Apr 2025

Relevant Coursework: NLP, Deep Learning, IOT, Data structures and Algorithms, Theory of Computation, Probability and Statistics, Linear Algebra

B.S. Data Science & Programming, Indian Institute of Technology, Madras (IIT-M)

Sep 2021 - Apr 2025

Relevant Coursework: Business Data Management, AI: Search Methods for Problem Solving, Statistics for Data Science, Design Thinking

EXPERIENCE

Cranium AI

Short Hills, NJ

AI Engineering Intern

Jun 2026 - Aug 2026

- Extended the platform's Public API by designing and shipping **10+ production endpoints** in .NET 10/C#, implementing cursor pagination and proxy resilience to boost data access and application integration.
- Implemented enterprise **RBAC features** with automated SQL migrations and query optimizations, empowering customers with granular access control while eliminating permission gaps across multi-tenant environments.
- Accelerated platform adoption and usability by **authoring an API Playbook and internal documentation**, while writing xUnit regression guards to ensure seamless data connector integration and zero-downtime releases.
- Strengthened platform reliability and testing by investigating multi-tenant authorization invariants and expanding xUnit/Moq test suites across 1,700+ CI/CD pipeline tests to prevent security regression.

TVS Credit

Chennai

Data Analytics Intern

Aug 2023 - Oct 2023

- Built an independent ML-based Customer Micro-Segmentation project, applying clustering algorithms to categorize **500,000 customers** by financial behavior for targeted marketing.
- Performed data cleaning, feature engineering, and exploratory analysis on large-scale customer financial data, resolving data-quality issues to improve model reliability.
- Evaluated and tuned clustering model performance across the project lifecycle, iterating on features and parameters to strengthen segment separation and accuracy.
- Partnered with data scientists to translate model output into analytical insights, guiding senior management on data-driven customer-targeting decisions.

PROJECTS

LLM-Based AI Tutor Chatbot

Nov 2025 - Dec 2025

Tech Stack: Python, LLMs, RAG, Vector Search, Knowledge Graphs, Speech-to-Text, Text-to-Speech

- Engineered a full-stack AI tutor chatbot integrating an **LLM** with a **Retrieval-Augmented Generation (RAG)** pipeline and a knowledge graph built from course materials.
- Built a complete voice interaction pipeline using **speech-to-text** and **text-to-speech** models, enabling natural, hands-free learning conversations.
- Designed context management and retrieval logic that grounded model responses in relevant course content, improving accuracy and personalization for each learner's query.

Deep Learning & YOLO-Inspired Plant Disease Detection with Smart Pesticide Recommendation

Jan 2025 - Apr 2025

Tech Stack: Python, YOLOv5, YOLOv8, OpenCV, PyTorch, Image Augmentation

- Investigated and benchmarked **YOLOv5** and **YOLOv8** architectures for plant disease detection across the PlantSeg and PlantDoc datasets.
- Curated and annotated a balanced **2,000-image** subset from PlantSeg with preprocessing, augmentation, and Gaussian blur, raising **mAP@50 to 0.849 (v5) and 0.864 (v8)**.
- Delivered **0.685 (v5) and 0.692 (v8) mAP@50** on PlantDoc, with YOLOv8 outperforming on precision and recall, translating results into actionable pesticide recommendations for agricultural risk mitigation.

Employing GRU for Identification of Fake & Authentic Tweets

Sep 2024 - Nov 2024

Tech Stack: Python, TensorFlow/PyTorch, GRU, NLP, Feature Engineering

- Developed a **GRU**-based sequence model for real-time detection of fake and authentic tweets, achieving up to **96.25% accuracy** across the PolitiFact, GossipCop, and PHEME datasets.
- Owned the full pipeline, data preprocessing, feature extraction, model training, and evaluation, to improve classification accuracy.
- Co-authored and presented "Employing Gated Recurrent Units for the Identification of Fake and Authentic Tweets" at **ICRTAC '24** and **ICCCNT '25 (publication in progress)**.

Smart Medicine Box

Jan 2024 - May 2024

Tech Stack: Python, TensorFlow/Keras, OpenCV, IoT, Mobile App Integration

- Developed an **IoT-enabled** image-classification system to identify medicines from a custom dataset of **800+ images** across **20+ medicines**.
- Curated and refined the training dataset, tuning the model to accurately recognize a diverse range of medicine packaging.
- Integrated real-time medication reminders through a companion mobile application to support timely, reliable dosing.

SKILLS

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLMs, RAG, Prompt Engineering, Sequence Modelling (GRU), Object Detection (YOLO), Model Training & Evaluation, Claude, Claude Code

Programming: Python, JavaScript, HTML, Java, C/C++, C#, SQL, SQLite, MongoDB, Spark, Hadoop

Tools & Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, Docker, Git, Bruno, API Development, Data Visualization (Power BI, D3.js)

LEADERSHIP & ACTIVITIES

Event Organizer, IOTHinc (IoT Club), Vellore Institute of Technology

- Led planning for the annual university fest, organizing a hackathon (**150 participants**) and an IoT-themed treasure hunt (**75 participants**).

Member, Society of Women Engineers (SWE), New York University

Secretary, NYU's Premier Indian Classical Dance Club

President, Annettes Club, Rotary Club of Madras Midtown